TL494 Pulse-Width-Modulation Control Circuits

1 Features

- Complete PWM power-control circuitry
- Uncommitted outputs for 200mA sink or source current
- Output control selects single-ended or push-pull operation
- Internal circuitry prohibits double pulse at either output
- Variable dead time provides control over total range
- Internal regulator provides a stable 5V reference supply with 5% tolerance
- Circuit architecture allows easy synchronization

2 Applications

- Desktop PCs
- Microwave ovens
- Power supplies: AC/DC, isolated, with or without PFC
- Server PSUs
- Solar micro-inverters
- Washing machines: low-end and high-end
- E-bikes
- Power: telecom/server AC/DC supplies: dual controller: analog
- Smoke detectors
- Solar power inverters

3 Description

The TL494 device incorporates all the functions required in the construction of a pulse-width-modulation (PWM) control circuit on a single chip. Designed primarily for power-supply control, this device offers the flexibility to tailor the power-supply control circuitry to a specific application.

The TL494 device contains two error amplifiers, an on-chip adjustable oscillator, a dead-time control (DTC) comparator, a pulse-steering control flip-flop, a 5V, 5%-precision regulator, and output-control circuits.

The error amplifiers exhibit a common-mode voltage range from –0.3V to V_{CC} – 2V. The dead-time control comparator has a fixed offset that provides approximately 5% dead time. The on-chip oscillator can be bypassed by terminating RT to the reference output and providing a sawtooth input to CT, or it can drive the common circuits in synchronous multiple-rail power supplies.

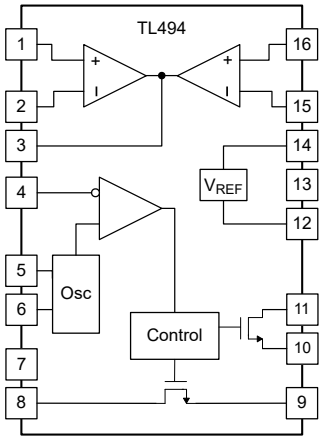
The uncommitted output transistors provide either common-source or source-follower output capability. The TL494 device provides for push-pull or single-ended output operation, which can be selected through the output-control function. The architecture of this device prohibits the possibility of either output being pulsed twice during push-pull operation.

The TL494 device is characterized for operation from 0°C to 70°C. The TL494I device is characterized for operation from –40°C to 85°C.

Package Information

PART NUMBER	PACKAGE(1)	PACKAGE SIZE(2)
TL494	D (SOIC, 16)	9.90mm × 3.91mm
	N (PDIP, 16)	19.30mm × 6.35mm
	NS (SOP, 16)	10.30mm × 5.30mm
	PW (TSSOP, 16)	5.00mm × 4.40mm

(1) For all available packages, see the orderable addendum at the end of the datasheet.
(2) The package size (length × width) is a nominal value and includes pins, where applicable.



Simplified Block Diagram

An IMPORTANT NOTICE at the end of this data sheet addresses availability, warranty, changes, use in safety-critical applications, intellectual property matters and other important disclaimers. PRODUCTION DATA.

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4 Pin Configuration and Functions

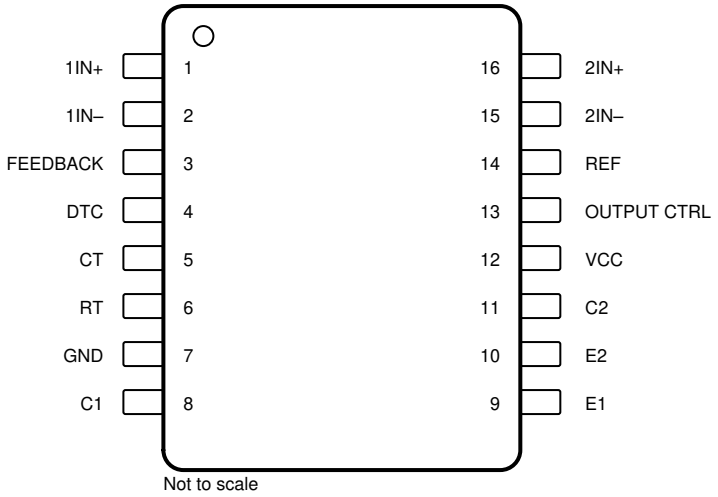


Figure 4-1. D, N, NS, or PW Package 16-Pin SOIC, PDIP, SO, or TSSOP Top View

Table 4-1. Pin Functions

PIN		TYPE(1)	DESCRIPTION
NAME	NO.		
1IN-	2	I	Inverting input to error amplifier 1
1IN+	1	I	Noninverting input to error amplifier 1
2IN-	15	I	Inverting input to error amplifier 2
2IN+	16	I	Noninverting input to error amplifier 2
C1	8	O	Drain terminal of MOSFET output 1
C2	11	O	Drain terminal of MOSFET output 2
CT	5	—	Capacitor terminal used to set oscillator frequency
DTC	4	I	Dead-time control comparator input
E1	9	O	Source terminal of MOSFET output 1
E2	10	O	Source terminal of MOSFET output 2
FEEDBACK	3	I	Input pin for feedback
GND	7	—	Ground
OUTPUT CTRL	13	I	Selects single-ended/parallel output or push-pull operation
REF	14	O	5V reference regulator output
RT	6	—	Resistor terminal used to set oscillator frequency
V_{CC}	12	—	Positive Supply

(1) I = input, O = output

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5 Specifications

5.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)(1)

		MIN	MAX	UNIT
V_{CC}	Supply voltage(2)		41	V
V_I	Amplifier input voltage		V_{CC} + 0.3	V
V_O	Collector output voltage		41	V
I_O	Collector output current		250	mA
	Lead temperature 1,6mm (1/16 inch) from case for 10 seconds		260	°C
T_{stg}	Storage temperature range	–65	150	°C

(1) Operation outside the *Absolute Maximum Ratings* may cause permanent device damage. *Absolute Maximum Ratings* do not imply functional operation of the device at these or any other conditions beyond those listed under *Recommended Operating Conditions*. If used outside the *Recommended Operating Conditions* but within the *Absolute Maximum Ratings*, the device may not be fully functional, and this may affect device reliability, functionality, performance, and shorten the device lifetime.
(2) All voltages are with respect to the network ground terminal.

5.2 ESD Ratings

			MAX	UNIT
$V_{(ESD)}$	Electrostatic discharge	Human body model (HBM), per ANSI/ESDA/JEDEC JS-001, all pins	500	V
		Charged device model (CDM), per JEDEC specification JESD22-C101, all pins	200	

5.3 Recommended Operating Conditions

			MIN	MAX	UNIT
V_{CC}	Supply voltage		7	40	V
V_I	Amplifier input voltage		–0.3	V_{CC} – 2	V
V_O	Collector output voltage			40	V
	Collector output current (each transistor)			200	mA
	Current into feedback terminal			0.3	mA
f_{OSC}	Oscillator frequency		1	300	kHz
C_T	Timing capacitor		0.47	10000	nF
R_T	Timing resistor		1.8	500	kΩ
T_A	Operating free-air temperature	TL494C	0	70	°C
		TL494I	–40	85	

5.4 Thermal Information

THERMAL METRIC(1)		TL494				UNIT
		D (SOIC)	N (PDIP)	NS (SO)	PW (TSSOP)	
		16 PINS	16 PINS	16 PINS	16 PINS	
$R_{\theta JA}$	Junction-to-ambient thermal resistance	88.0	73.8	95.3	114.0	°C/W
$R_{\theta JC(top)}$	Junction-to-case (top) thermal resistance	51.4	52.9	56.8	53.2	°C/W
$R_{\theta JB}$	Junction-to-board thermal resistance	50.6	49.2	62.0	71.8	°C/W
ψ_{JT}	Junction-to-top characterization parameter	14.5	31.3	19.6	7.3	°C/W
ψ_{JB}	Junction-to-board characterization parameter	50.1	48.6	61.3	71.2	°C/W

(1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC Package Thermal Metrics* application note.

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5.5 Electrical Characteristics, Reference Section

over recommended operating free-air temperature range, V_{CC} = 15V, f = 10kHz (unless otherwise noted)

PARAMETER	TEST CONDITIONS[1]	TL494C, TL494I			UNIT
		MIN	TYP[2]	MAX	
Output voltage (REF)	I_O = 1mA	4.75	5	5.25	V
Input regulation	V_{CC} = 7V to 40V		2	25	mV
Output regulation	I_O = 1mA to 10mA		6	15	mV
Output voltage change with temperature	ΔT_A = MIN to MAX		2	10	mV/V
Short-circuit output current[3]	REF = 0V		35		mA

(1) For conditions shown as MIN or MAX, use the appropriate value specified under recommended operating conditions.
(2) All typical values, except for parameter changes with temperature, are at T_A = 25°C.
(3) Duration of short circuit should not exceed one second.

5.6 Electrical Characteristics, Oscillator Section

C_T = 0.01µF, R_T = 12kΩ (see Figure 6-1)

PARAMETER	TEST CONDITIONS[1]	TL494C, TL494I			UNIT
		MIN	TYP[2]	MAX	
Frequency			10		kHz
Standard deviation of frequency[3]	All values of V_{CC}, C_T, R_T, and T_A constant		100		Hz/kHz
Frequency change with voltage	V_{CC} = 7V to 40V, T_A = 25°C		1		Hz/kHz
Frequency change with temperature[4]	ΔT_A = MIN to MAX			50	Hz/kHz

(1) For conditions shown as MIN or MAX, use the appropriate value specified under recommended operating conditions.
(2) All typical values, except for parameter changes with temperature, are at T_A = 25°C.
(3) Standard deviation is a measure of the statistical distribution about the mean as derived from the formula:

$$\sigma = \sqrt{\frac{\sum_{n=1}^{N} [x_n - \overline{X}]^2}{N-1}} \quad (1)$$

(4) Temperature coefficient of timing capacitor and timing resistor are not taken into account.

5.7 Electrical Characteristics, Error-Amplifier Section

See Figure 6-2

PARAMETER	TEST CONDITIONS	TL494C, TL494I			UNIT
		MIN	TYP[1]	MAX	
Input offset voltage	V_O (FEEDBACK) = 2.5V		2	10	mV
Input offset current	V_O (FEEDBACK) = 2.5V		25	250	nA
Input bias current	V_O (FEEDBACK) = 2.5V		0.2	1	µA
Common-mode input voltage range	V_{CC} = 7V to 40V	–0.3 to V_{CC} – 2			V
Open-loop voltage amplification	ΔV_O = 3V, V_O = 0.5V to 3.5V, R_L = 2kΩ	70	80		dB
Unity-gain bandwidth	V_O = 0.5V to 3.5V, R_L = 2kΩ		800		kHz
Common-mode rejection ratio	ΔV_O = 40V, T_A = 25°C	65	100		dB
Output sink current (FEEDBACK)	V_{ID} = –15mV to –5V, V (FEEDBACK) = 0.7V	0.3	0.7		mA
Output source current (FEEDBACK)	V_{ID} = 15mV to 5V, V (FEEDBACK) = 3.5V	–2			mA

(1) All typical values, except for parameter changes with temperature, are at T_A = 25°C.

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5.8 Electrical Characteristics, Output Section

PARAMETER		TEST CONDITIONS	MIN	TYP(1)	MAX	UNIT
C1, C2 off-state current		V_{CE} = 40V, V_{CC} = 40V		4	100	µA
E1,E2 off-state current		V_{CC} = V_C = 40V, V_E = 0			–100	µA
C1E1, C2E2 saturation voltage	Common source	V_E = 0, I_C = 200mA		0.5	1.3	V
	Source follower	$V_{O(C1 \text{ or } C2)}$ = 15V, I_E = –200mA		1.65	2.5	
Output control input current		V_I = V_{ref}			3.5	mA

(1) All typical values, except for temperature coefficient, are at T_A = 25°C.

5.9 Electrical Characteristics, Dead-Time Control Section

See Figure 6-1

PARAMETER	TEST CONDITIONS	MIN	TYP(1)	MAX	UNIT
Input bias current (DEAD-TIME CTRL)	V_I = 0 to 5.25V		–0.006	–10	µA
Maximum duty cycle, each output	V_I (DEAD-TIME CTRL) = 0, C_T = 0.01µF, R_T = 12kΩ		45%		—
Input threshold voltage (DEAD-TIME CTRL)	Zero duty cycle		2.5	3.3	V
	Maximum duty cycle	0			

(1) All typical values, except for temperature coefficient, are at T_A = 25°C.

5.10 Electrical Characteristics, PWM Comparator Section

See Figure 6-1

PARAMETER	TEST CONDITIONS	MIN	TYP(1)	MAX	UNIT
Input threshold voltage (FEEDBACK)	Zero duty cyle		3.4	4.5	V
Input sink current (FEEDBACK)	V (FEEDBACK) = 0.7V	0.3	0.7		mA

(1) All typical values, except for temperature coefficient, are at T_A = 25°C.

5.11 Electrical Characteristics, Total Device

PARAMETER	TEST CONDITIONS		MIN	TYP(1)	MAX	UNIT
Standby supply current	R_T = V_{ref}, All other inputs and outputs open	V_{CC} = 15V		1.62	10	mA
		V_{CC} = 40V		1.68	15	
Average supply current	V_I (DEAD-TIME CTRL) = 2V, See Figure 6-1			2.2		mA

(1) All typical values, except for temperature coefficient, are at T_A = 25°C.

5.12 Switching Characteristics

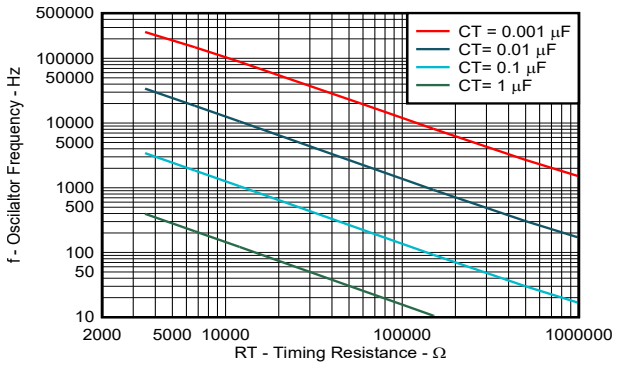
T_A = 25°C

PARAMETER	TEST CONDITIONS	MIN	TYP(1)	MAX	UNIT
Rise time	Common-source configuration, See Figure 6-3		70	200	ns
Fall time			20	100	ns
Rise time	Source-follower configuration, See Figure 6-4		85	200	ns
Fall time			35	100	ns

(1) All typical values, except for temperature coefficient, are at T_A = 25°C.

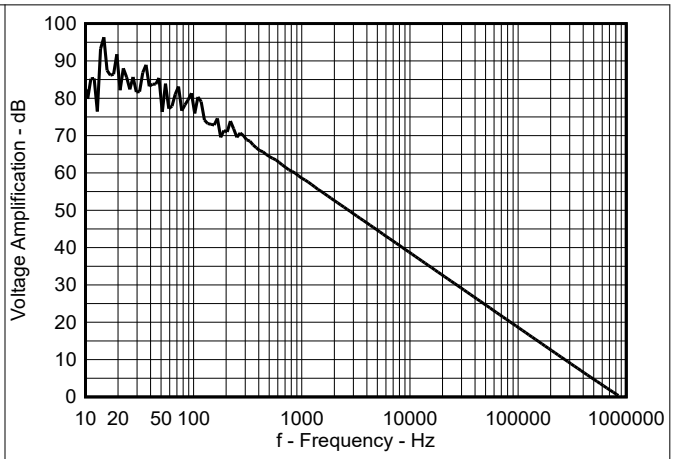
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5.13 Typical Characteristics



V_{CC} = 15V T_A = 25°C

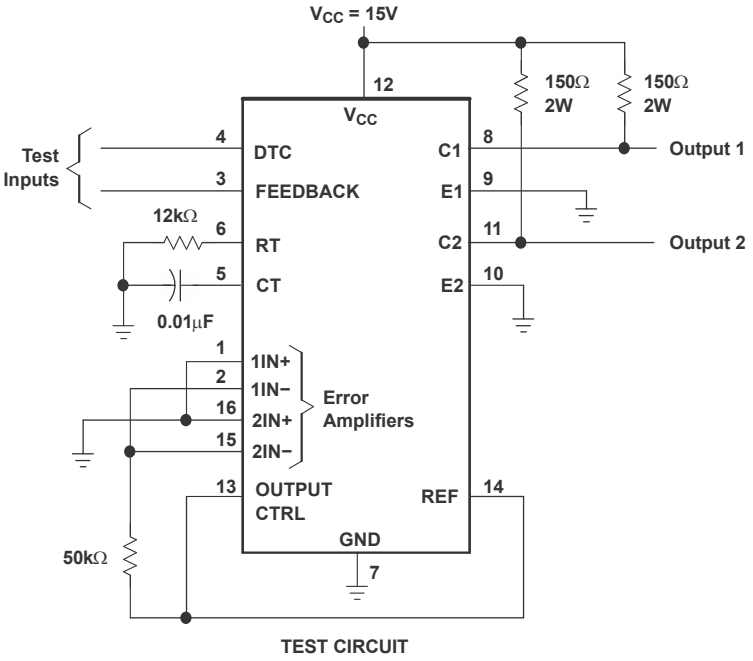
Figure 5-1. Oscillator Frequency vs Timing Resistance



V_{CC} = 15V T_A = 25°C ΔV_O = 3V

Figure 5-2. Amplifier Voltage Amplification vs Frequency

6 Parameter Measurement Information



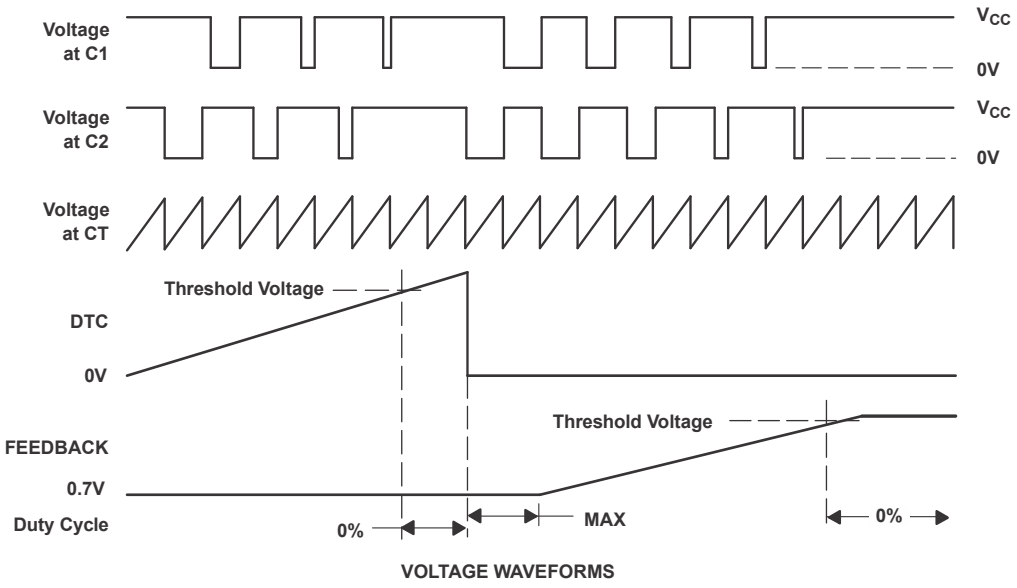


Figure 6-1. Operational Test Circuit and Waveforms

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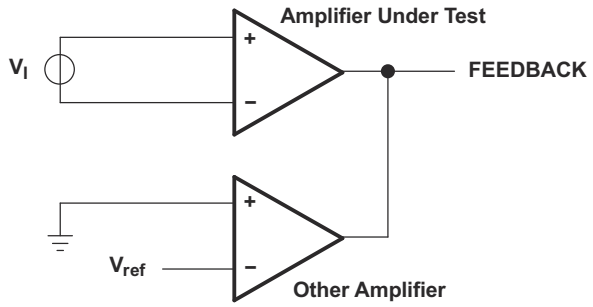


Figure 6-2. Amplifier Characteristics

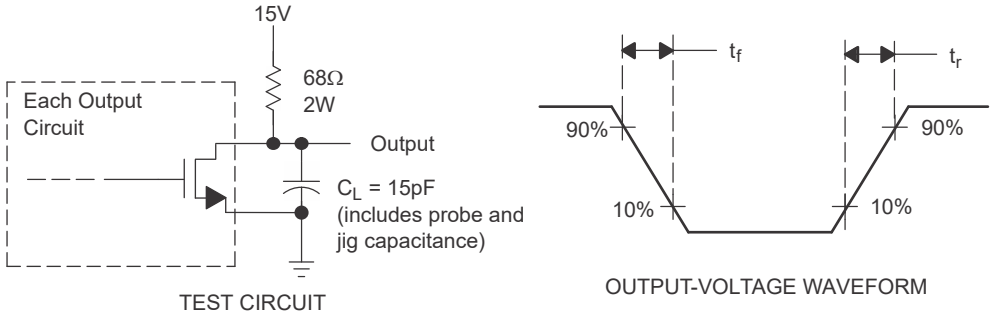


Figure 6-3. Common-Emitter Configuration

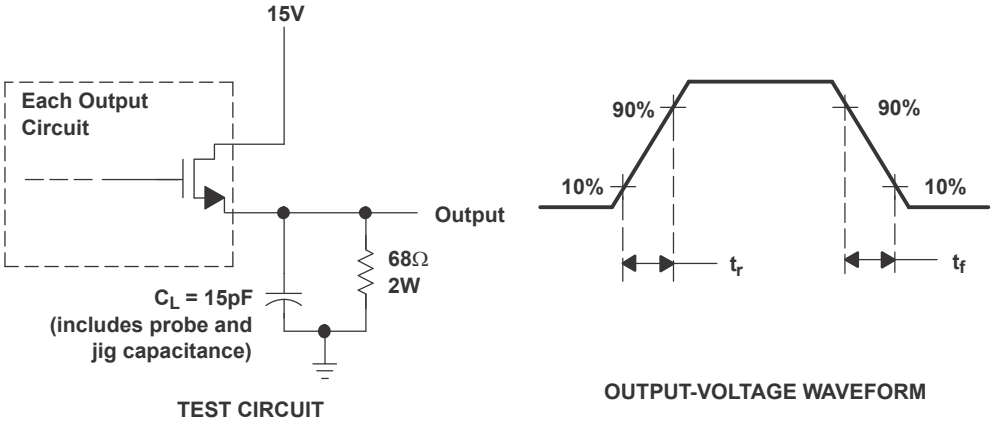


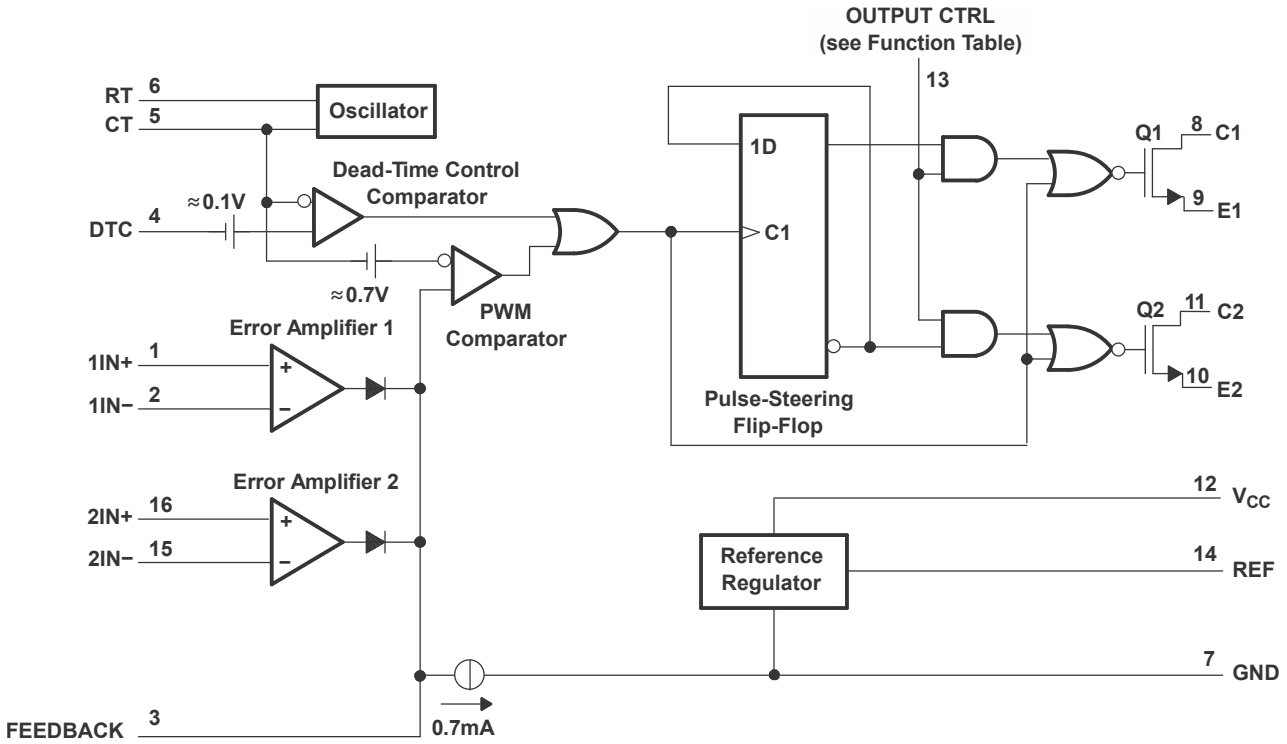
Figure 6-4. Emitter-Follower Configuration

7 Detailed Description

7.1 Overview

The design of the TL494 incorporates the primary building blocks required to control a switching power supply and addresses basic problems and reduces the amount of additional circuitry required in the total design. The TL494 is a fixed-frequency pulse-width-modulation (PWM) control circuit. The internal oscillator on the timing capacitor (CT) creates the sawtooth waveform. Achieve modulation of output pulses by comparing the sawtooth waveform to either of two control signals. The output stage is enabled during the time when the sawtooth voltage is greater than the voltage control signals. As the control signal increases, the time during which the sawtooth input is greater decreases; therefore, the output pulse duration decreases. A pulse-steering flip-flop alternately directs the modulated pulse to each of the two output transistors. For more information on the operation of the TL494, see the application notes located on ti.com.

7.2 Functional Block Diagram



7.3 Feature Description

7.3.1 5V Reference Regulator

The TL494 internal 5V reference regulator output is the REF pin. In addition to providing a stable reference, TL494 acts as a preregulator and establishes a stable supply to power the output-control logic, pulse-steering flip-flop, oscillator, dead-time control comparator, and PWM comparator. The regulator employs a band-gap circuit as the primary reference to maintain thermal stability of less than 100mV variation over the operating free-air temperature range from 0°C to 70°C. Short-circuit protection is provided to protect the internal reference and preregulator; 10mA of load current is available for additional bias circuits. The reference is internally programmed to an initial accuracy of ±5% and maintains a stability of less than 25mV variation over an input voltage range from 7V to 40V. For input voltages less than 7V, regulator output remains at 0V until the internal references reach a valid level, after which it transitions to 5V, resulting in more deterministic startup behavior as shown in Figure 8-8

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7.3.2 Oscillator

The oscillator provides a positive sawtooth waveform to the dead-time and PWM comparators for comparison to the various control signals.

Program the frequency of the oscillator by selecting timing components R_T and C_T. The oscillator charges the external timing capacitor, C_T, with a constant current. The external timing resistor, R_T determines the constant current value and produces a linear-ramp voltage waveform. When the voltage across C_T reaches 3V, the oscillator circuit discharges it, and the charging cycle reinitiates. Equation 2 determines the charging current.

$$I_{CHARGE} = \frac{3V}{R_T} \tag{2}$$

The period of the sawtooth waveform is:

$$T = \frac{3V \times C_T}{I_{CHARGE}} \tag{3}$$

The frequency of the oscillator becomes:

$$f_{OSC} = \frac{1}{R_T \times C_T} \tag{4}$$

However, the oscillator frequency is equal to the output frequency only for single-ended applications. For push-pull applications, the output frequency is one-half the oscillator frequency.

Single-ended applications:

$$f = \frac{1}{R_T \times C_T} \tag{5}$$

Push-pull applications:

$$f = \frac{1}{2R_T \times C_T} \tag{6}$$

7.3.3 Dead-time Control

The dead-time control input provides control of the minimum dead time (off time). The output of the comparator inhibits switching transistors Q1 and Q2 when the voltage at the input is greater than the ramp voltage of the oscillator. An internal offset of 110mV confirms a minimum dead time of approximately 3% with the dead time control input grounded. Applying a voltage to the dead-time control input can impose additional dead time. This provides a linear control of the dead time from its minimum of 3% to 100% as the input voltage varies from 0V to 3.3V, respectively. With full-range control, control the output from external sources without disrupting the error amplifiers. The dead-time control input is a relatively high-impedance input (I_I < 10µA) and must be used where additional control of the output duty cycle is required. However, for proper control, terminate the input. An open circuit is an undefined condition.

7.3.4 Comparator

The comparator is biased from the 5V reference regulator, providing isolation from the input supply for improved stability. The input of the comparator does not exhibit hysteresis, so protection against false triggering near the threshold must be provided. The comparator has a response time of 400ns from either of the control-signal inputs to the output transistors, with only 100mV of overdrive. This establishes positive control of the output within one-half cycle for operation within the recommended 300kHz range.

7.3.5 Pulse-Width Modulation (PWM)

The comparator also provides modulation control of the output pulse width. For this, the ramp voltage across timing capacitor C_T is compared to the control signal present at the output of the error amplifiers. The timing capacitor input incorporates a series diode that is omitted from the control signal input. This requires the control

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signal (error amplifier output) to be ≅ 0.7V greater than the voltage across C_T to inhibit the output logic, and confirms maximum duty-cycle operation without requiring the control voltage to sink to a true ground potential. The output pulse width varies from 97% of the period to 0 as the voltage present at the error amplifier output varies from 0.5V to 3.5V, respectively.

7.3.6 Error Amplifiers

Both high-gain error amplifiers receive bias from the V_I supply rail. This permits a common-mode input voltage range from –0.3V to 2V less than V_I. Both amplifiers behave characteristically of a single-ended single-supply amplifier, in that each output is active high only. This allows each amplifier to pull up independently for a decreasing output pulse-width demand. With both outputs ORed together at the inverting input node of the PWM comparator, the amplifier demanding the minimum pulse out dominates. The amplifier outputs are biased low by a current sink to provide maximum pulse width out when both amplifiers are biased off.

7.3.7 Output-Control Input

The output-control input determines whether the output transistors operate in parallel or push-pull. This input is the supply source for the pulse-steering flip-flop. The output-control input is asynchronous and has direct control over the output, independent of the oscillator or pulse-steering flip-flop. The input condition is intended to be a fixed condition that the application defines. For parallel operation, ground the output-control input. This disables the pulse-steering flip-flop and inhibits its outputs. In this mode, both output transistors in parallel transmit the pulses seen at the output of the dead-time control/PWM comparator. For push-pull operation, connect the output-control input to the internal 5V reference regulator. Under this condition, each of the output transistors is enabled, alternately, by the pulse-steering flip-flop.

7.3.8 Output Transistors

Two output transistors are available on the TL494. Both transistors are configured as open drain/open source, and each can sink or source up to 200mA. The transistors have a saturation voltage of less than 1.3V in the common-source configuration and less than 2.5V in the source-follower configuration. The outputs are protected against excessive power dissipation to prevent damage and do not employ sufficient current limiting to allow them to operate as current-source outputs.

7.4 Device Functional Modes

When the OUTPUT CTRL pin is tied to ground, the TL494 is operating in single-ended or parallel mode. When the OUTPUT CTRL pin is tied to V_{REF}, the TL494 is operating in normal push-pull operation.

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8 Application and Implementation

Note

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes, as well as validating and testing their design implementation to confirm system functionality.

8.1 Application Information

The following design example uses the TL494 to create a 5V/10A power supply. For more information, see Designing Switching Voltage Regulators With the TL494.

8.2 Typical Application

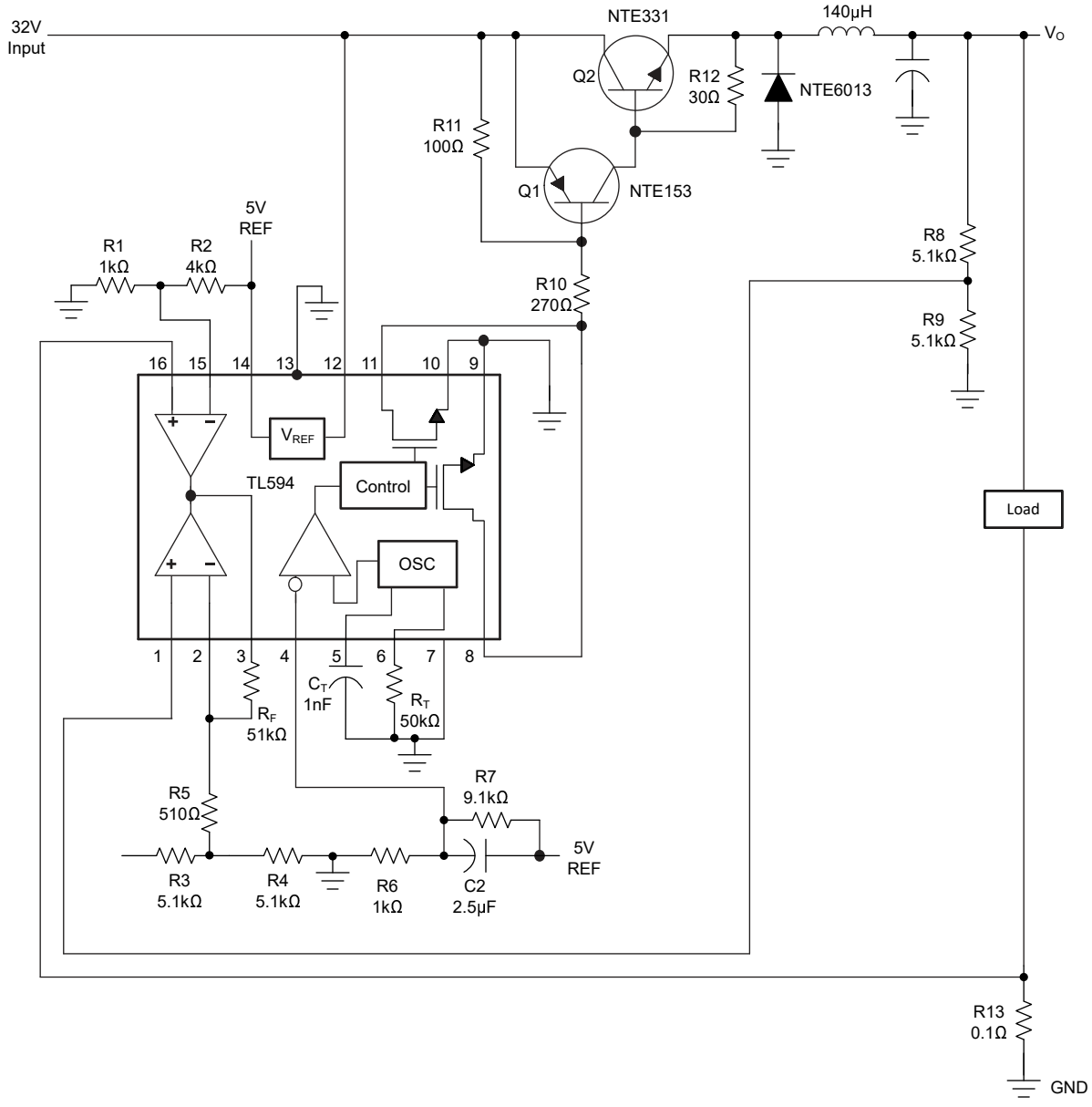


Figure 8-1. Switching and Control Sections

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8.2.1 Design Requirements

- V_I = 32V
- V_O = 5V
- I_O = 10A
- f_{OSC} = 20kHz switching frequency
- V_R = 20mV peak-to-peak (V_{RIPPLE})
- ΔI_L = 1.5A inductor current change

8.2.2 Detailed Design Procedure

8.2.2.1 Input Power Source

The 32V DC power source for this supply uses a 120V input, 24V output transformer rated at 75VA. The 24V secondary winding feeds a full-wave bridge rectifier, followed by a current-limiting resistor (0.3Ω) and two filter capacitors (see Figure 8-2).

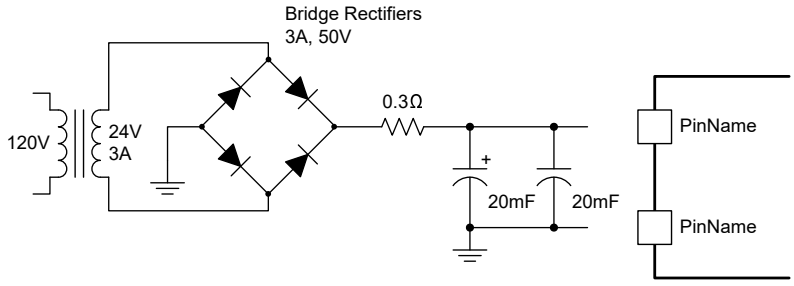


Figure 8-2. Input Power Source

Equation 7 and Equation 8 determine the output current and voltage.

$$V_{RECTIFIER} = V_{SECONDARY} \times \sqrt{2} = 24V \times \sqrt{2} = 34V \quad (7)$$

$$I_{RECTIFIER[AVG]} \cong \frac{V_O}{V_I} \times I_O \cong \frac{5V}{32V} \times 10A = 1.6A \quad (8)$$

The 3A/50V full-wave bridge rectifier meets these calculated conditions. Figure 8-1 shows the switching and control sections.

8.2.2.2 Control Circuits

8.2.2.2.1 Oscillator

Connecting an external capacitor and resistor to pins 5 and 6 controls the TL494 oscillator frequency. The oscillator is set to operate at 20kHz, using the component values calculated by Equation 4 and Equation 9:

Select C_T = 0.001µF and calculate R_T:

$$R_T = \frac{1}{f_{OSC} \times C_T} = \frac{1}{\left[20 \times 10^3\right] \times \left[0.001 \times 10^{-6}\right]} = 50k\Omega \quad (9)$$

8.2.2.2.2 Error Amplifier

The error amplifier compares a sample of the 5V output to the reference and adjusts the PWM to maintain a constant output current (see Figure 8-3).

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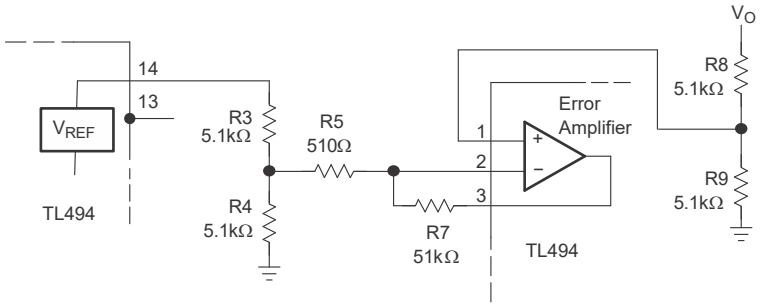


Figure 8-3. Error-Amplifier Section

The TL494 internal 5V reference is divided to 2.5V by R3 and R4. The output-voltage error signal also is divided to 2.5V by R8 and R9. If the output must be regulated to exactly 5.0V, use a 10kΩ potentiometer in place of R8 to provide an adjustment.

To increase the stability of the error-amplifier circuit, the output of the error amplifier is fed back to the inverting input through R_T, reducing the gain to 101.

8.2.2.2.3 Current-Limiting Amplifier

The power supply id designed for a 10A load current and an I_L swing of 1.5A, therefore, the short-circuit current is:

$$I_{SC} = I_O + \frac{I_L}{2} = 10.75A \tag{10}$$

Figure 8-4 shows the current-limiting circuit.

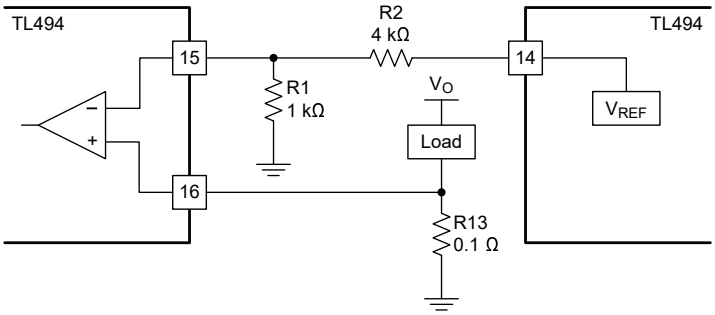


Figure 8-4. Current-Limiting Circuit

Resistors R1 and R2 set the reference of approximately 1V on the inverting input of the current-limiting amplifier. Resistor R13, in series with the load, applies 1 V to the noninverting terminal of the current-limiting amplifier when the load current reaches 10A. The output pulse duration reduces accordingly. Equation 11 calculate the value of R13.

$$R13 = \frac{1V}{10A} = 0.1\Omega \tag{11}$$

8.2.2.2.4 Soft Start and Dead Time

To reduce stress on the switching transistors at the start-up time, reduce the start-up surge that occurs as the output filter capacitor charges. The availability of the dead-time control makes implementation of a soft-start circuit relatively simple (see Figure 8-5).

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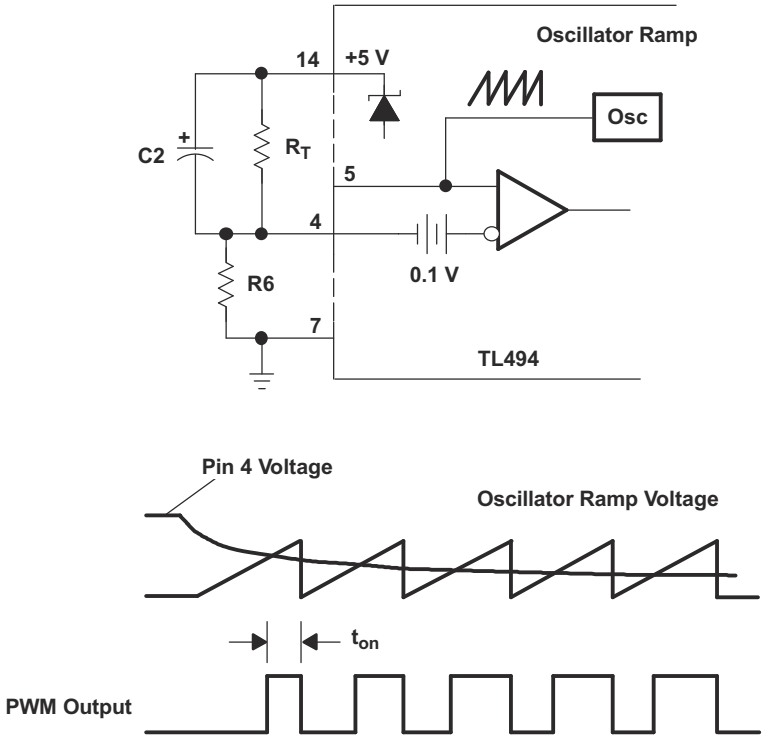


Figure 8-5. Soft-Start Circuit

The soft-start circuit allows the pulse duration at the output to increase slowly (see Figure 8-5) by applying a negative slope waveform to the dead-time control input (pin 4).

Initially, capacitor C2 forces the dead-time control input to follow the 5V regulator, which disables the outputs (100% dead time). As the capacitor charges through R6, the output pulse duration slowly increases until the control loop takes command. With a resistor ratio of 1:10 for R6 and R7, the voltage at pin 4 after start-up is 0.1 × 5V, or 0.5V.

The soft-start time generally is in the range from 25 to 100 clock cycles. If selecting 50 clock cycles at a 20kHz switching rate, the soft-start time is:

$$t = \frac{1}{f} = \frac{1}{20\text{kHz}} = 50\mu\text{s per clock cycle} \tag{12}$$

Calculate the value of the capacitor using:

$$C2 = \frac{\text{soft} - \text{start time}}{R6} = \frac{50\mu\text{s} \times 50 \text{ cycles}}{1\text{k}\Omega} = 2.5\mu\text{F} \tag{13}$$

This helps eliminate any false signals created while applying power to the control circuit.

8.2.2.3 Inductor Calculations

Figure 8-6 shows the switching circuit.

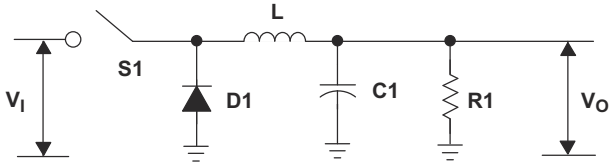


Figure 8-6. Switching Circuit

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The size of the inductor (L) required is:

$$d = \text{duty cycle} = \frac{V_O}{V_I} = \frac{5V}{32V} = 0.156 \quad (14)$$

$$f = 20\text{kHz (design objective)} \quad (15)$$

$$t_{ON} = \text{time on (S1 closed)} = \left(\frac{1}{f}\right) \times d = 7.8\mu s \quad (16)$$

$$t_{OFF} = \text{time off (S1 open)} = \left(\frac{1}{f}\right) - \text{ton} = 42.2\mu s \quad (17)$$

$$\begin{aligned} L &\approx (V_I - V_O) \times \frac{t_{ON}}{\Delta I_L} \\ &\approx \frac{[(32V - 5V) \times 7.8\mu s]}{1.5A} \\ &\approx 140.4\mu H \end{aligned} \quad (18)$$

8.2.2.4 Output Capacitance Calculations

After calculating the filter inductor, calculate the value of the output filter capacitor to meet the output ripple requirements. Model an electrolytic capacitor as a series connection of an inductance, a resistance, and a capacitance. To provide good filtering, the ripple frequency must be far below the frequencies at which the series inductance becomes important. So, the two components of interest are the capacitance and the effective series resistance (ESR). Calculate the maximum ESR according to the relation between the specified peak-to-peak ripple voltage and the peak-to-peak ripple current.

$$ESR[\text{max}] = \frac{\Delta V_{O[\text{ripple}]}}{\Delta I_L} = \frac{V}{1.5A} \cong 0.067\Omega \quad (19)$$

Use Equation 20 to calculate the minimum capacitance of C3 necessary to maintain the V_O ripple voltage at less than the 100mV design objective.

$$C3 = \frac{\Delta I_L}{8f\Delta V_O} = \frac{1.5A}{8 \times 20 \times 10^3 \times 0.1V} = 94\mu F \quad (20)$$

A 220mF, 60V capacitor is selected because the 60V capacitor has a maximum ESR of 0.074Ω and a maximum ripple current of 2.8A.

8.2.2.5 Transistor Power-Switch Calculations

The transistor power switch is constructed with an NTE153 pnp drive transistor and an NTE331 npn output transistor. These two power devices connect in a pnp hybrid Darlington circuit configuration (see Figure 8-7).

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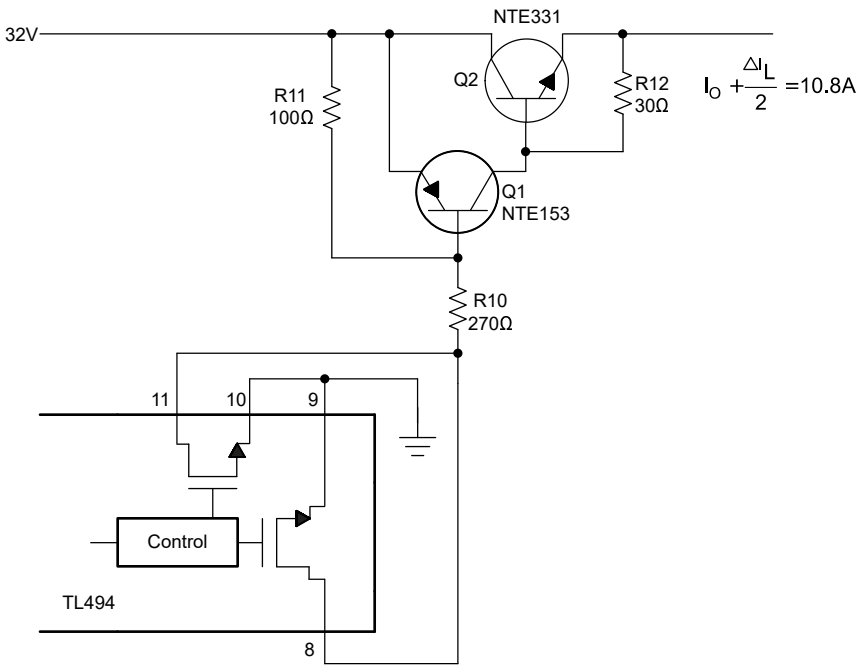


Figure 8-7. Power-Switch Section

The hybrid Darlington circuit must be saturated at a maximum output current of I_O + $\Delta I_L/2$ or 10.8 A. The Darlington h_{FE} at 10.8 A must be high enough not to exceed the 250mA maximum output collector current of the TL494. Based on published NTE153 and NTE331 specifications, calcualte the required power-switch minimum drive using Equation 21 through Equation 23 to be 144mA:

$$h_{FE}[Q1] \text{ at } I_C \text{ of } 3A = 15 \tag{21}$$

$$h_{FE}[Q2] \text{ at } I_C \text{ of } 10.0A = 5 \tag{22}$$

$$i_B \geq \frac{I_O + \frac{I_L}{2}}{h_{FE}[Q2] \times h_{FE}[Q1]} \geq 144mA \tag{23}$$

Calculate the value of R10 using:

$$R10 \leq \frac{V_I - [V_{BE}(Q1) + V_{CE}(TL494)]}{i_B} = \frac{32 - [1.5 + 0.7]}{0.144}$$
$$R10 \leq 207\Omega \tag{24}$$

Based on these calculations, the nearest standard resistor value of 220Ω is selected for R10. Resistors R11 and R12 permit the discharge of carriers in switching transistors when turned off.

The power supply described demonstrates the flexibility of the TL494 PWM control circuit. This power-supply design demonstrates many of the power-supply control methods the TL494 provides, as well as the versatility of the control circuit.

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8.2.3 Application Curves for Output Characteristics

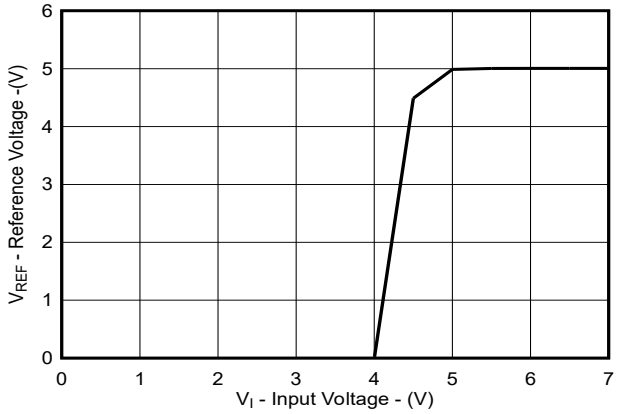


Figure 8-8. Reference Voltage vs Input Voltage

8.3 Power Supply Recommendations

The TL494 is designed to operate from an input voltage supply range from 7V to 40V. Well-regulate the input supply. If the input supply is located more than a few inches from the device, additional bulk capacitance can be required, in addition to the ceramic bypass capacitors. A tantalum capacitor with a value of 47µF is a typical choice, however this can vary depending upon the delivery of the output power.

8.4 Layout

8.4.1 Layout Guidelines

Always try to use a low EMI inductor with a ferrite type closed core, for example, toroid and encased E core inductors. Only use open core if they have low EMI characteristics and are located a bit more away from the low-power traces and components. Make the poles perpendicular to the PCB as well if using an open core. Stick cores usually emit the most unwanted noise.

8.4.1.1 Feedback Traces

Try to run the feedback trace as far from the inductor and noisy power traces as possible. Confirm that the feedback trace is as direct as possible and somewhat thick. These two sometimes involve a trade-off, but keeping it away from inductor EMI and other noise sources is the more critical of the two. Run the feedback trace on the side of the PCB opposite of the inductor with a ground plane separating the two.

8.4.1.2 Input/Output Capacitors

When using a low value ceramic input filter capacitor, locate the capacitor as close to the VCC pin of the IC as possible to eliminate as many trace inductance effects as possible and give the internal IC rail a cleaner voltage supply. Some designs require the use of a feed-forward capacitor connected from the output to the feedback pin as well, usually for stability reasons. In this case, position the capacitor as close to the IC as possible. Using surface-mount capacitors also reduces lead length and lessens the chance of noise coupling into the effective antenna created by through-hole components.

8.4.1.3 Compensation Components

For stability, place external compensation components close to the IC. Surface mount components are recommended here as well for the same reasons discussed for the filter capacitors. Do not locate surface-mount components very close to the inductor either.

8.4.1.4 Traces and Ground Planes

- Make all of the power (high current) traces as short, direct, and thick as possible. It is good practice on a standard PCB board to make the traces an absolute minimum of 15mils (0.381mm) per ampere.
- Confirm that the inductor, output capacitors, and output diode are as close to each other possible. This helps reduce the EMI radiated by the power traces due to the high switching currents through them and reduces

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lead inductance and resistance as well, which in turn reduces noise spikes, ringing, and resistive losses that produce voltage errors.

- Connect the grounds of the IC, input capacitors, output capacitors, and output diode (if applicable) close together directly to a ground plane. Have a ground plane on both sides of the PCB to reduce noise by reducing ground loop errors and by absorbing more of the EMI radiated by the inductor.
- For multi-layer boards with more than two layers, use a ground plane to separate the power plane (where the power traces and components are) and the signal plane (where the feedback and compensation and components are) for improved performance.
- On multi-layer boards using vias is required to connect traces and different planes.
- It is good practice to use one standard via per 200mA of current if the trace must conduct a significant amount of current from one plane to the other.
- Arrange the components so that the switching current loops curl in the same direction. Due to the way switching regulators operate, there are two power states. One state when the switch is on and one when the switch is off. During each state there is a current loop made by the power components that are currently conducting. Place the power components so that during each of the two states the current loop is conducting in the same direction. This prevents magnetic field reversal caused by the traces between the two half-cycles and reduces radiated EMI.

8.4.2 Layout Example

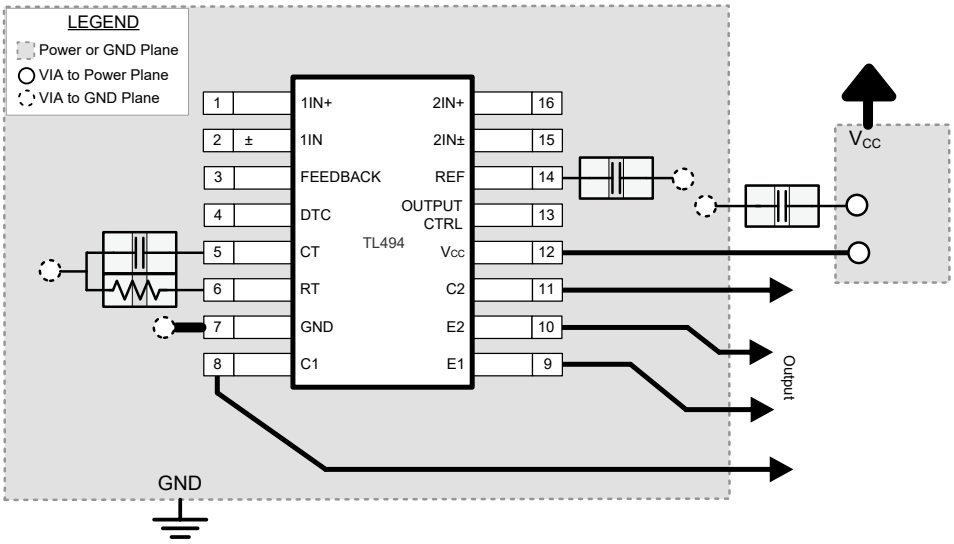


Figure 8-9. Operational Amplifier Board Layout for Noninverting Configuration

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9 Device and Documentation Support

9.1 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on ti.com. Click on *Notifications* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

9.2 Support Resources

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9.3 Documentation Support

9.3.1 Related Documentation

- Texas Instruments, *Designing Switching Voltage Regulators With the TL494* application note
- Texas Instruments, *Semiconductor and IC Package Thermal Metrics* application note

9.4 Trademarks

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This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

9.6 Glossary

TI Glossary	This glossary lists and explains terms, acronyms, and definitions.

10 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Changes from Revision I (July 2022) to Revision J (September 2026)	Page
• Updated the numbering format for tables, figures, and cross-references throughout the document	1
• Updated output transistor description and symbols from BJT to MOSFET throughout the document	1
• Changed values in the *Thermal Information* table for all packages	4
• Changed output regulation typical value from 1mV to 6mV	5
• Changed short-circuit output current typical value from 25mA to 35mA	5
• Changed frequency change with temperature max value from 10Hz/kHz to 50Hz/kHz	5
• Changed open-loop voltage amplification typical value from 95dB to 80dB	5
• Changed common-mode rejection ratio typical value from 80dB to 100dB	5
• Changed output C1, C2 off-state current typical value from 2µA to 4µA	6
• Changed output C1E1, C2E2 saturation voltage in common source configuration typical value from 1.1V to 0.5V	6
• Changed output C1E1, C2E2 saturation voltage in source follower configuration typical value from 1.5V to 1.65V	6
• Changed input bias current (DEAD-TIME CTRL) typical value from -2µA to -0.006µA	6
• Changed input threshold voltage (DEAD-TIME CTRL) typical value from 3V to 2.5V	6
• Changed input threshold voltage (FEEDBACK) typical value from 4V to 3.4V	6

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- Changed standby supply current for VCC = 15V typical value from 6mA to 1.62mA....6
- Changed standby supply current for VCC = 40V typical value from 9mA to 1.68mA....6
- Changed average supply current typical value from 7.5mA to 2.2mA....6
- Changed rise time in common source configuration typical value from 100ns to 70ns....6
- Changed fall time in common source configuration typical value from 25ns to 20ns....6
- Changed rise time in source follower configuration typical value from 100ns to 85ns....6
- Changed fall time in source follower configuration typical value from 40ns to 35ns....6
- Updated Oscillator Frequency vs Timing Resistance, Amplifier Voltage Amplification vs Frequency, and removed others....7
- Updated description to align VREF behavior changes for input voltages less than 7V....10
- Updated the formatting for inductor calculations....16
- Updated reference voltage vs input voltage application curve....19

Changes from Revision H (March 2017) to Revision I (July 2022) **Page**

- Updated the numbering format for tables, figures, and cross-references throughout the document....1

11 Mechanical, Packaging, and Orderable Information

The following pages include mechanical packaging and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser based versions of this data sheet, refer to the left hand navigation.

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PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package | Pins	Package qty | Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
TL494CD	Obsolete	Production	SOIC (D) | 16	-	-	Call TI	Call TI	0 to 70	TL494C
TL494CDR	Active	Production	SOIC (D) | 16	2500 | LARGE T&R	Yes	NIPDAU | SN	Level-1-260C-UNLIM	0 to 70	TL494C
TL494CDR.A	Active	Production	SOIC (D) | 16	2500 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	TL494C
TL494CDRG4	Obsolete	Production	SOIC (D) | 16	-	-	Call TI	Call TI	0 to 70	TL494C
TL494CN	Active	Production	PDIP (N) | 16	25 | TUBE	Yes	NIPDAU	N/A for Pkg Type	0 to 70	TL494CN
TL494CN.A	Active	Production	PDIP (N) | 16	25 | TUBE	Yes	NIPDAU	N/A for Pkg Type	0 to 70	TL494CN
TL494CNE4	Active	Production	PDIP (N) | 16	25 | TUBE	Yes	NIPDAU	N/A for Pkg Type	0 to 70	TL494CN
TL494CNSR	Active	Production	SOP (NS) | 16	2000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	TL494
TL494CNSR.A	Active	Production	SOP (NS) | 16	2000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	TL494
TL494CNSRG4	Active	Production	SOP (NS) | 16	2000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	TL494
TL494CPW	Obsolete	Production	TSSOP (PW) | 16	-	-	Call TI	Call TI	0 to 70	T494
TL494CPWR	Active	Production	TSSOP (PW) | 16	2000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	T494
TL494CPWR.A	Active	Production	TSSOP (PW) | 16	2000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	T494
TL494ID	Obsolete	Production	SOIC (D) | 16	-	-	Call TI	Call TI	-40 to 85	TL494I
TL494IDR	Active	Production	SOIC (D) | 16	2500 | LARGE T&R	Yes	NIPDAU | SN	Level-1-260C-UNLIM	-40 to 85	TL494I
TL494IDR.A	Active	Production	SOIC (D) | 16	2500 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	TL494I
TL494IDRG4	Obsolete	Production	SOIC (D) | 16	-	-	Call TI	Call TI	-40 to 85	TL494I
TL494IN	NRND	Production	PDIP (N) | 16	25 | TUBE	Yes	NIPDAU	N/A for Pkg Type	-40 to 85	TL494IN
TL494IN.A	NRND	Production	PDIP (N) | 16	25 | TUBE	Yes	NIPDAU	N/A for Pkg Type	-40 to 85	TL494IN
TL494INE4	NRND	Production	PDIP (N) | 16	25 | TUBE	Yes	NIPDAU	N/A for Pkg Type	-40 to 85	TL494IN

(1) **Status:** For more details on status, see our product life cycle.

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) **RoHS values:** Yes, No, RoHS Exempt. See the TI RoHS Statement for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

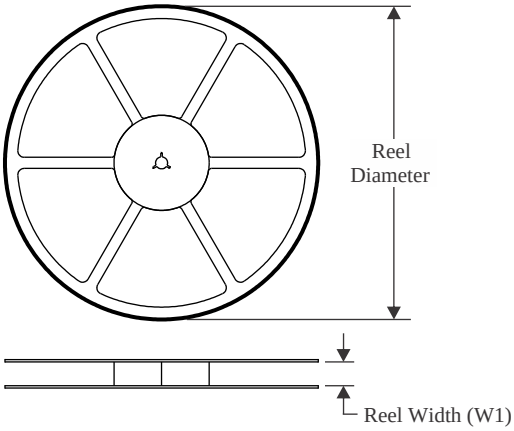
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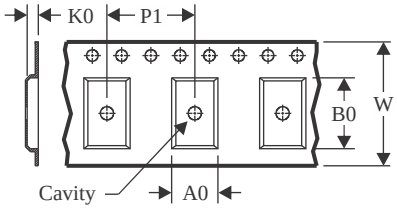
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TAPE AND REEL INFORMATION

REEL DIMENSIONS

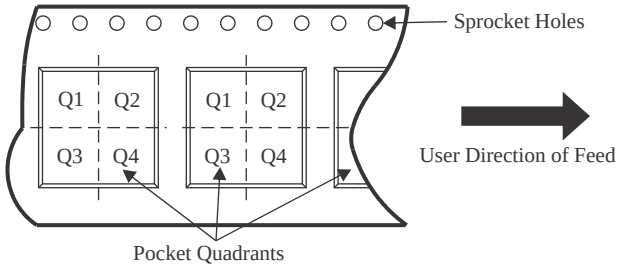


TAPE DIMENSIONS



A0	Dimension designed to accommodate the component width
B0	Dimension designed to accommodate the component length
K0	Dimension designed to accommodate the component thickness
W	Overall width of the carrier tape
P1	Pitch between successive cavity centers

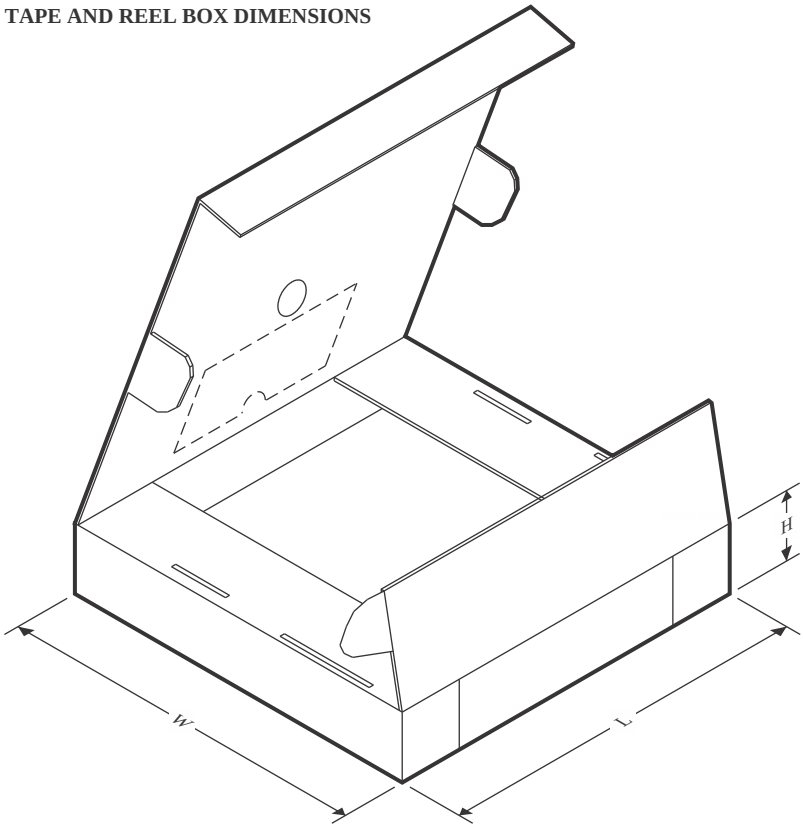
QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
TL494CDR	SOIC	D	16	2500	330.0	16.4	6.5	10.3	2.1	8.0	16.0	Q1
TL494CNSR	SOP	NS	16	2000	330.0	16.4	8.45	10.55	2.5	12.0	16.2	Q1
TL494CPWR	TSSOP	PW	16	2000	330.0	12.4	6.9	5.6	1.6	8.0	12.0	Q1
TL494IDR	SOIC	D	16	2500	330.0	16.4	6.5	10.3	2.1	8.0	16.0	Q1

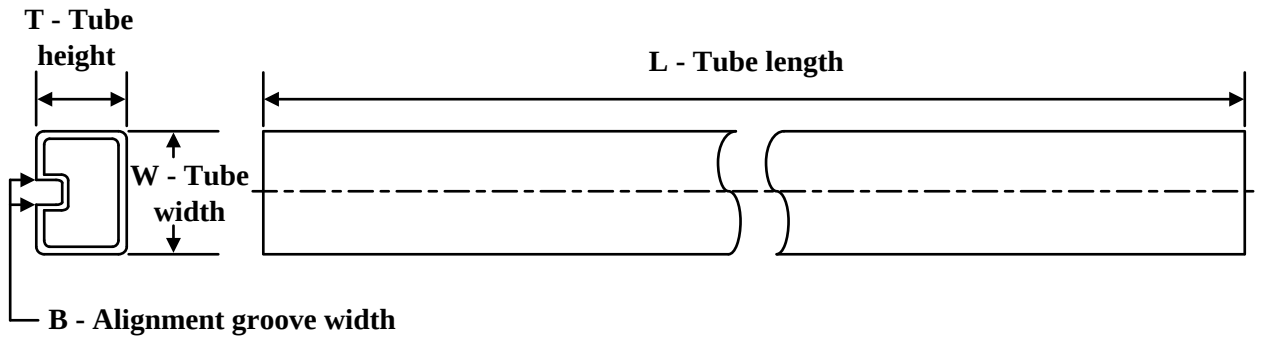
TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
TL494CDR	SOIC	D	16	2500	353.0	353.0	32.0
TL494CNSR	SOP	NS	16	2000	353.0	353.0	32.0
TL494CPWR	TSSOP	PW	16	2000	353.0	353.0	32.0
TL494IDR	SOIC	D	16	2500	353.0	353.0	32.0

TUBE



*All dimensions are nominal

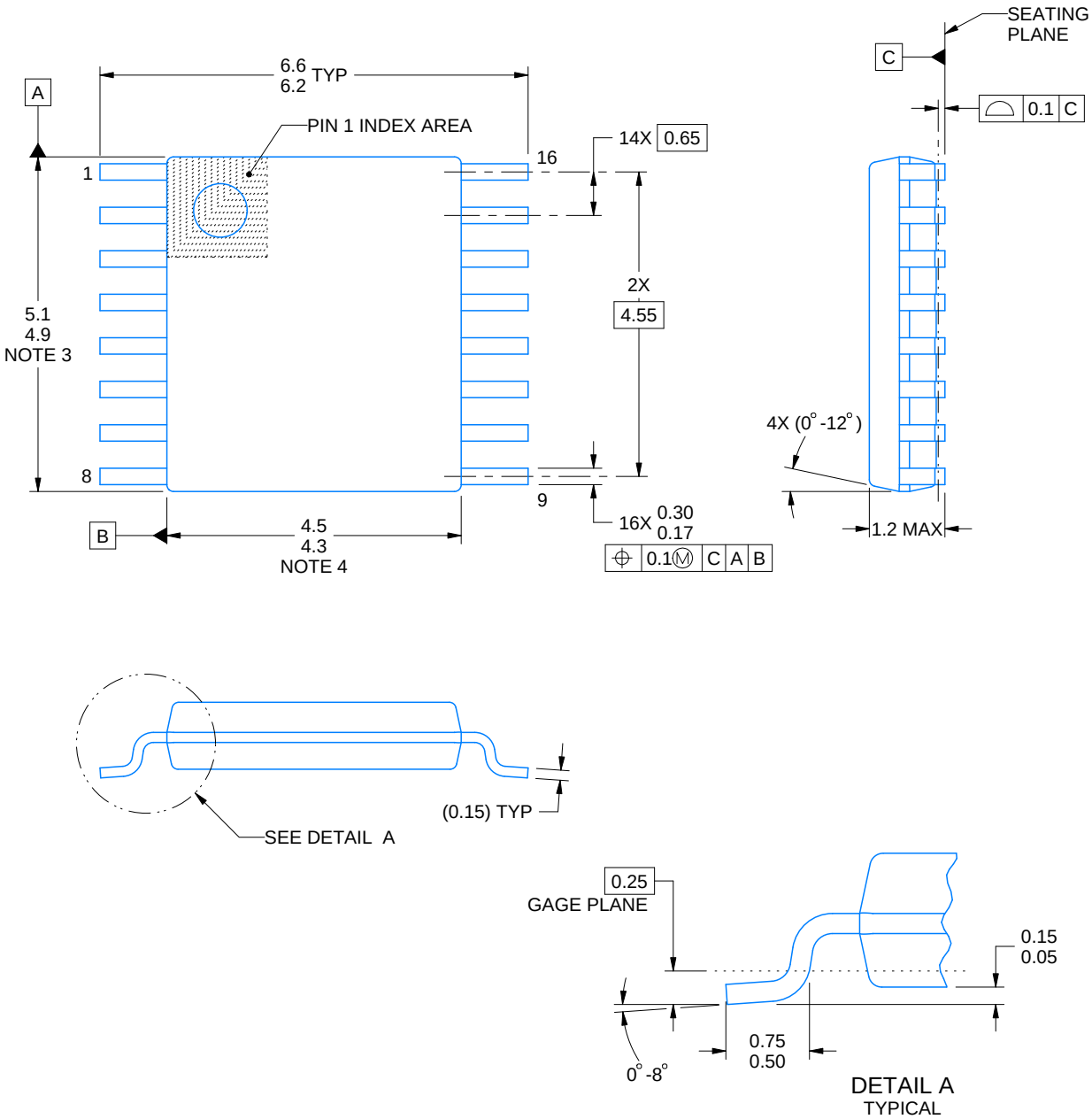
Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (µm)	B (mm)
TL494CN	N	PDIP	16	25	506	13.97	11230	4.32
TL494CN	N	PDIP	16	25	506	13.97	11230	4.32
TL494CN.A	N	PDIP	16	25	506	13.97	11230	4.32
TL494CN.A	N	PDIP	16	25	506	13.97	11230	4.32
TL494CNE4	N	PDIP	16	25	506	13.97	11230	4.32
TL494CNE4	N	PDIP	16	25	506	13.97	11230	4.32
TL494IN	N	PDIP	16	25	506	13.97	11230	4.32
TL494IN	N	PDIP	16	25	506	13.97	11230	4.32
TL494IN.A	N	PDIP	16	25	506	13.97	11230	4.32
TL494IN.A	N	PDIP	16	25	506	13.97	11230	4.32
TL494INE4	N	PDIP	16	25	506	13.97	11230	4.32
TL494INE4	N	PDIP	16	25	506	13.97	11230	4.32

PACKAGE OUTLINE

PW0016A

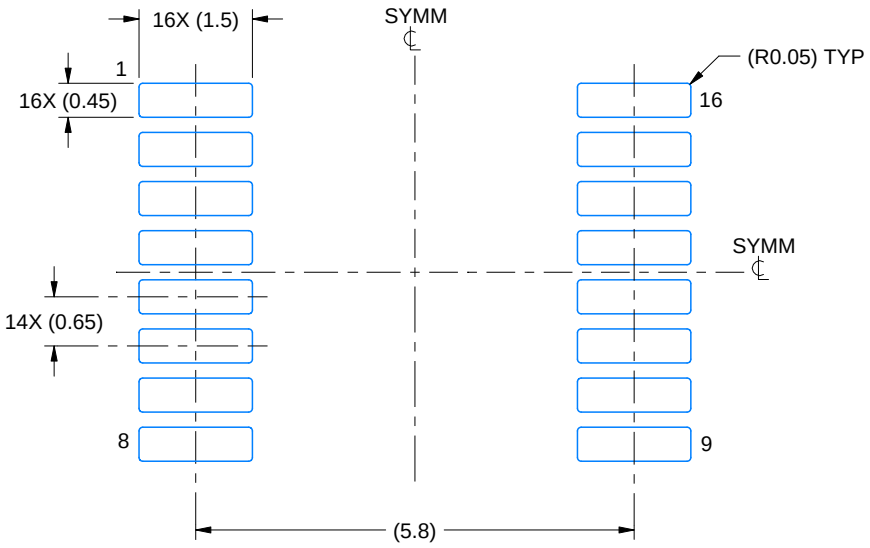
TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE

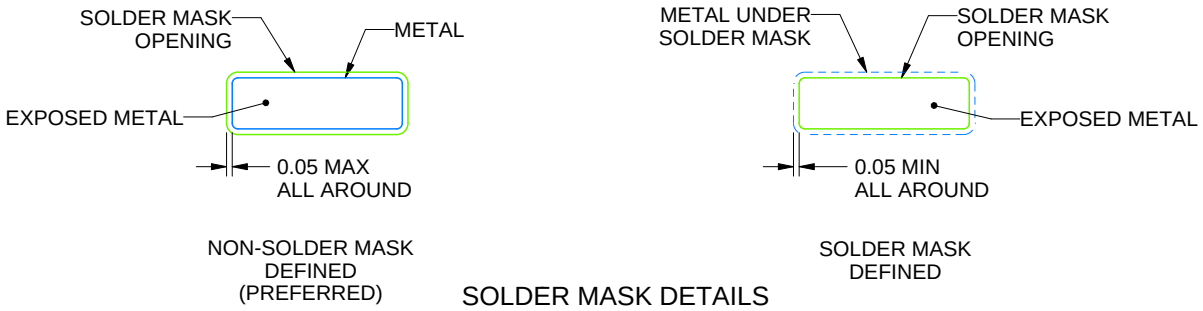


NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.
5. Reference JEDEC registration MO-153.



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE: 10X

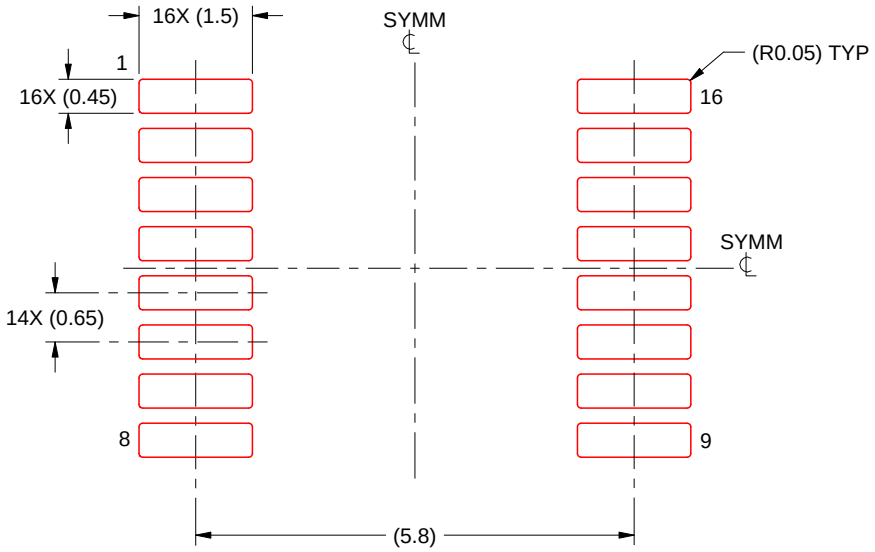


SOLDER MASK DETAILS

4220204/B 12/2023

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.



SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE: 10X

4220204/B 12/2023

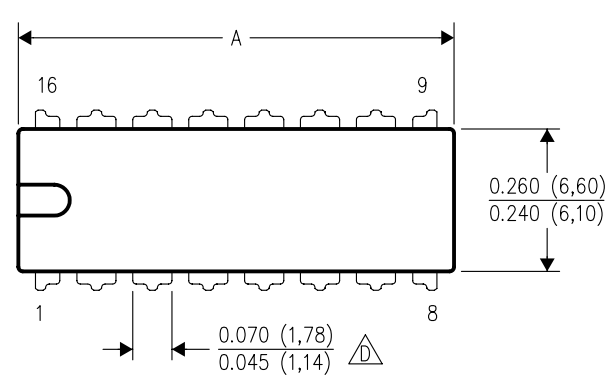
NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

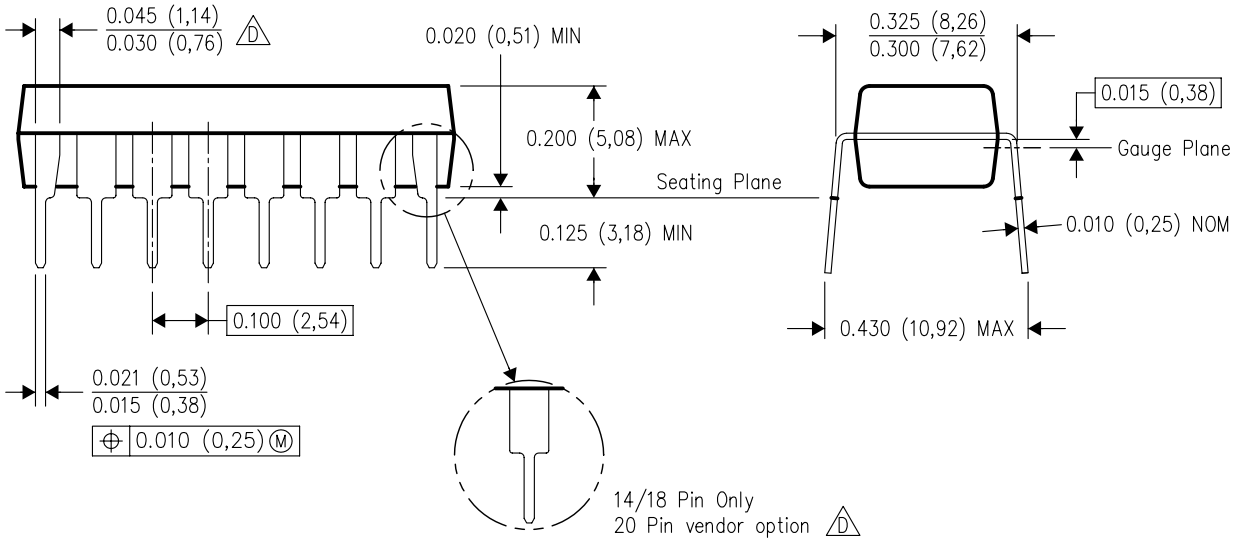
N (R–PDIP–T**)

PLASTIC DUAL–IN–LINE PACKAGE

16 PINS SHOWN

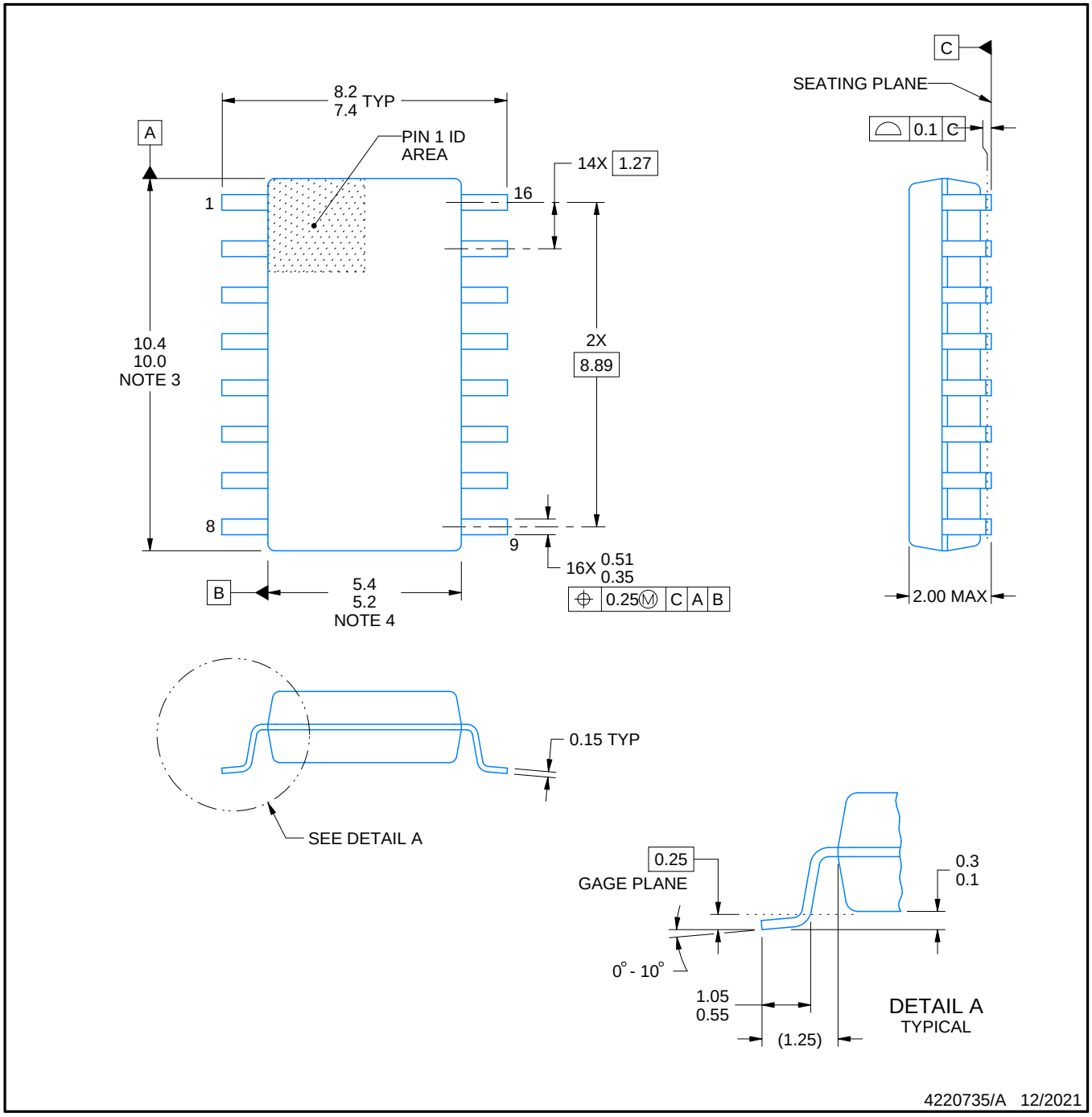


DIM \ PINS **	14	16	18	20
A MAX	0.775 (19,69)	0.775 (19,69)	0.920 (23,37)	1.060 (26,92)
A MIN	0.745 (18,92)	0.745 (18,92)	0.850 (21,59)	0.940 (23,88)
MS–001 VARIATION (C)	AA	BB	AC	AD



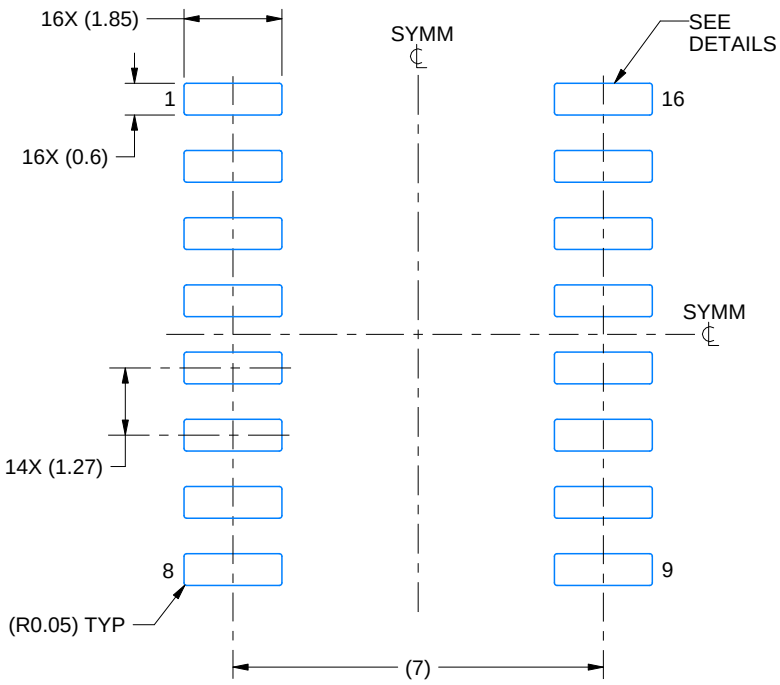
4040049/E 12/2002

NOTES:
A. All linear dimensions are in inches (millimeters).
B. This drawing is subject to change without notice.
C. Falls within JEDEC MS–001, except 18 and 20 pin minimum body length (Dim A).
D. The 20 pin end lead shoulder width is a vendor option, either half or full width.

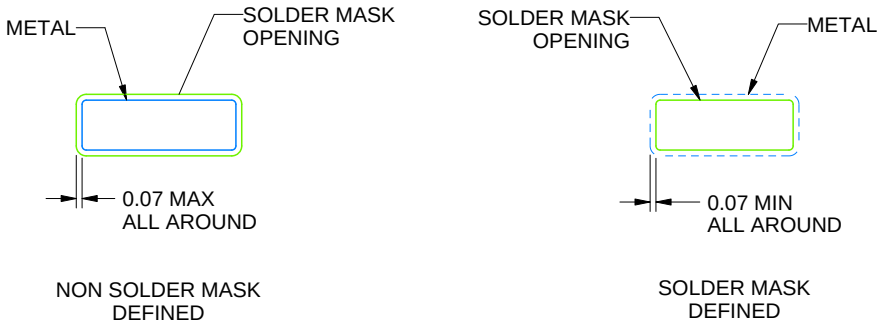


NOTES:

1. All linear dimensions are in millimeters. Dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm, per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm, per side.



LAND PATTERN EXAMPLE
SCALE:7X

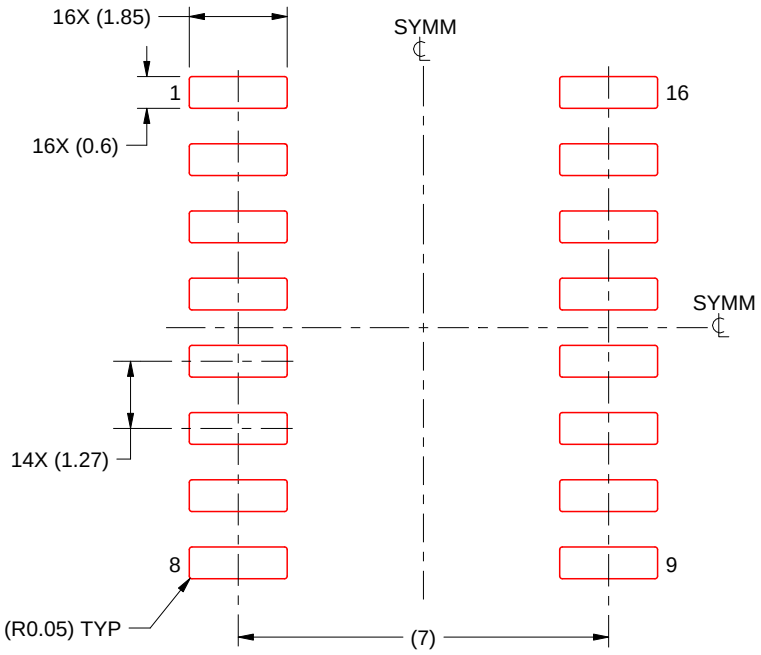


SOLDER MASK DETAILS

4220735/A 12/2021

NOTES: (continued)

5. Publication IPC-7351 may have alternate designs.
6. Solder mask tolerances between and around signal pads can vary based on board fabrication site.



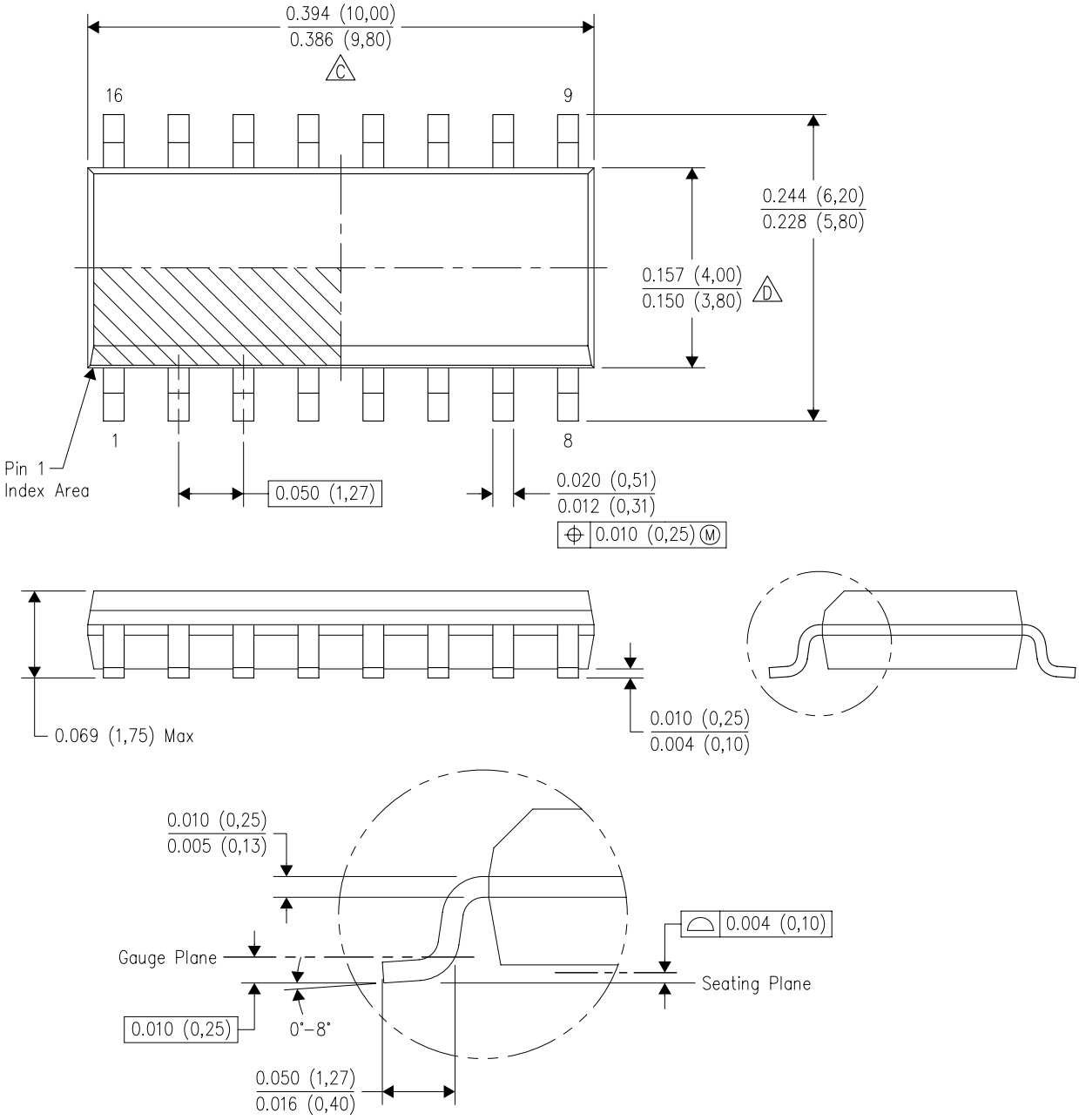
SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE:7X

4220735/A 12/2021

NOTES: (continued)

7. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
8. Board assembly site may have different recommendations for stencil design.

D (R–PDSO–G16) PLASTIC SMALL OUTLINE



NOTES:
A. All linear dimensions are in inches (millimeters).
B. This drawing is subject to change without notice.
C. Body length does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.006 (0,15) each side.
D. Body width does not include interlead flash. Interlead flash shall not exceed 0.017 (0,43) each side.
E. Reference JEDEC MS–012 variation AC.

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